

Over the past 75 years the study of kaon physics has been integral to the development of the standard model. Since 2016, the NA62 experiment has been the leading experiment in the field of kaon physics. However, with the NA62 experiment scheduled up to CERN Long Shutdown 3 (LS3), which is expected in autumn 2026, it is important to investigate future experiments. It is then required to investigate the future of kaon physics and the potential experiments that could succeed NA62. This chapter will discuss the potential future of kaon physics, the experiments that could be used to further the field, and the work I contributed to these experiments.

0.1 High Intensity Kaon Experiments (HIKE)

High-intensity Kaon Experiments (HIKE) is a proposed project to continue kaon physics at CERN in the ECN3 experimental hall and represented a long-phased program after LS3. HIKE would profit from a four to six-fold increase in beam intensity compared to NA62 and included cutting-edge detector technologies that will be described in this section. The beam intensity requirements for HIKE are described in Table 1. To accommodate the higher intensities, upgrades are needed to the infrastructure that provided the beam to ECN3. A number of working groups such, as the ECN3 beam delivery group within the Physics Beyond Colliders community at CERN [1], was already conducting efforts to consolidate the work required to upgrade the infrastructure. Three phases are foreseen within the HIKE program, the first being a continuation of NA62 charged kaon physics with the high-intensity beam and detector upgrades to accommodate the rate increase and the goal of measuring the branching ratio of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ to a 5 % precision. The second phase would involve neutral kaon physics with charged secondary particle tracking with the goal to measure the branching ratio of $K_L \rightarrow \pi^0 l^+ l^-$ with $l = e, \mu$. The third phase is designed to observe the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay and measure its branching ratio to a 20 % precision.

HIKE Phase	Protons on Target/Spill	K Decays/Year
Phase 1 (K^+)	1.3×10^{13}	2×10^{13}
Phase 2 (K_L + tracking)	2×10^{13}	3.8×10^{13}
Phase 3 (K_L , KLEVER)	2×10^{13}	1.3×10^{13}

Table 1: Intensity requirements for three phases of HIKE. The number of kaon decays per year corresponds to those that decay within the HIKE fiducial volume.

0.1.1 Phase 1

The experimental design of NA62 has been proven as a successful setup for the precision measurement of the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay, described in Section ?? . The HIKE Phase 1 detector setup, shown in Figure 1, is similar to the NA62 setup described in Section ?? with new and upgraded detectors. The HIKE detectors are required to withstand a four times greater particle rate with respect to NA62 equating to 1.3×10^{13} protons per spill on the target. The expected HIKE Phase 1 beam had similar characteristics to the NA62 one, with a 75 GeV/ c secondary hadron beam produced from a 400 GeV/ c primary proton beam from the SPS accelerator onto a target at an angle of 0 rad.

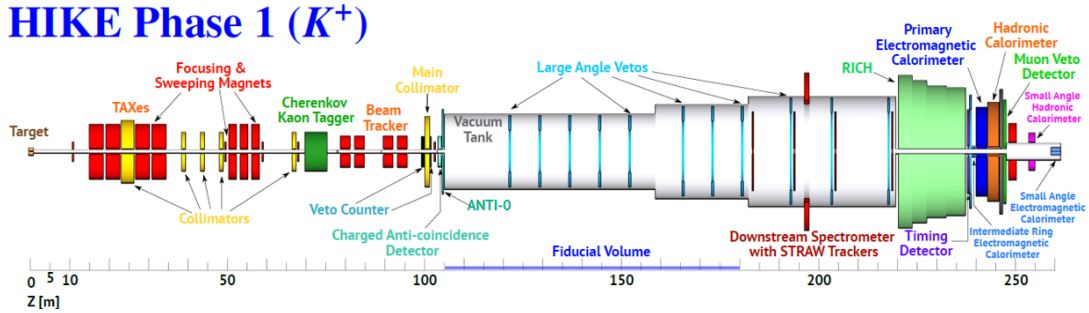


Figure 1: Diagram of HIKE Phase 1 detector setup [2].

The detectors before the fiducial volume included an upgraded photon-detection system for the KTAG, described in Section 0.3, and a new GTK chip design. It is proposed to upgrade the existing NA62 Vetocounter with scintillating fibre (SciFi) technologies to improve the granularity and to accommodate the increased beam intensity.

Major upgrades are envisioned to the LAV and STRAW detectors within the

fiducial volume. A number of LAV stations are to be upgraded to a new lead scintillator design, similar to the Vacuum Veto System (VVS) planned for the CKM experiment [3, 4]. The LAV detectors are crucial to the subsequent phases of HIKE, especially Phase 3, and are designed to fulfil the requirements of the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ measurement even when used in Phase 1. The STRAW spectrometer is to use an upgraded a novel 5 mm straw design with a layout shown in Figure 2. The reduction in the straw diameter would decrease drift times while increasing granularity and resulting in an expected improvement in the track angle and momentum resolution of 10-20 % compared to NA62.

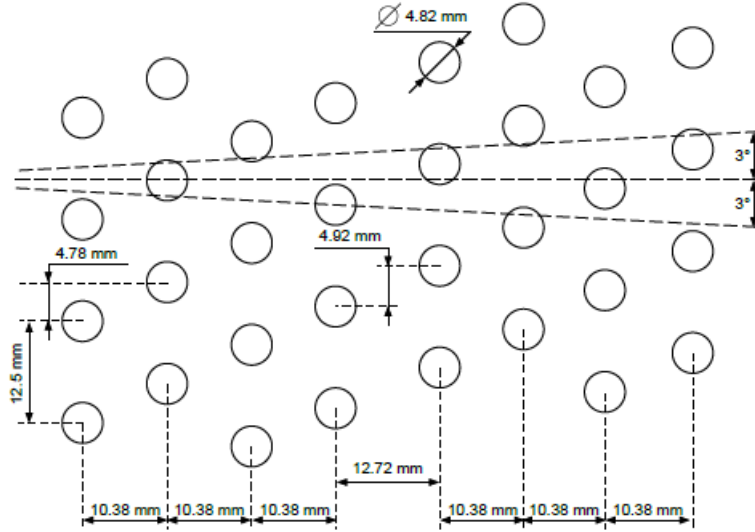


Figure 2: Diagram of the new STRAW spectrometer layout with 5 mm straw tubes [2].

Downstream of the fiducial volume, the PMT arrays of the RICH detector are upgraded, as described in Section 0.4. The HIKE timing detector would profit from the current NA62-CHOD detector with upgraded Photomultiplier Tubes (PMTs), which are to be chosen from the outcome of studies from other detectors such as KTAG, RICH and MUVs. For HIKE Phase 1, the LKr is proposed as the primary electromagnetic calorimeter given its excellent energy resolution, but the required timing resolution is to be achieved by upgrading its readout system. A number of other detectors are to be upgraded or replaced with higher granularity, such as the hadronic calorimeter, MUVs, IRC, and SAC.

It is expected that HIKE Phase 1 would run between LS3 and LS4, with the exact

years being undetermined. About 2.5×10^{-6} kaon decays in the fiducial volume is expected per proton on target during HIKE phase 1. Given the expected protons per spill of 1.3×10^{13} (four times NA62) and a nominal year of 200 data-taking days with an average of 3000 spills per day, would result in an expected 2×10^{13} kaon decays per year foreseen during HIKE phase 1. As a consequence the Phase 1 goals could be completed in 5 years of data taking.

0.1.2 Phase 2

HIKE Phase 2 is a multi-purpose neutral K_L decay experiment with the primary goal being to measure the branching ratios of the $K_L \rightarrow \pi^0 l^+ l^-$ decays. The detector setup for phase 2, shown in Figure 3, would involve minimal modification to the experimental setup of Phase 1, with the removal of detectors that are only applicable to the charged kaon beam. The KTAG, GTK, RICH, and SAC detectors are to be removed, and the STRAW spectrometer would be shortened to account for a softer momentum of the secondary particles. The STRAW chambers would be reconfigured to have central beam holes to account for the neutral beam. The MNP33 magnet requires a reduction in strength by 20% to account for the shorter spectrometer, resulting in a momentum kick of 210 MeV/c. The fiducial volume would be increased to 90 m compared to 75 m in Phase 1.

HIKE Phase 2 (K_L)

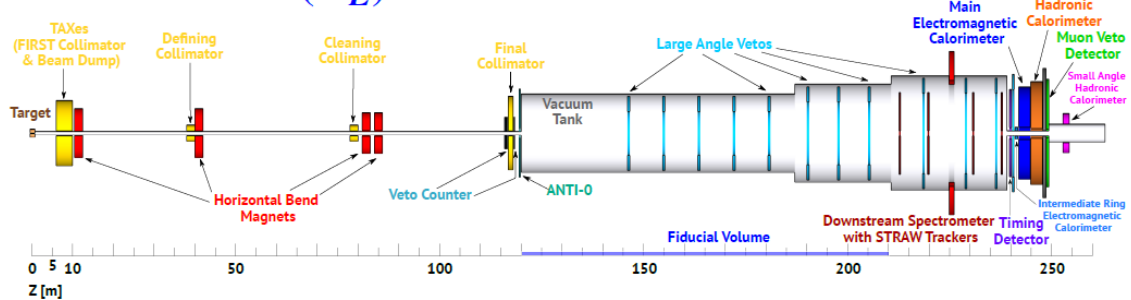


Figure 3: Diagram of HIKE Phase 2 detector setup [2].

The design proposed a production angle of 2.4 mrad on the target producing an expected K_L yield of 5.4×10^{-5} per proton. Assuming an integrated proton flux of 1.2×10^{19} protons on target per year would result in 3.8×10^{13} K_L decays in the fiducial volume per year. The mean beam momentum at the entry of the fiducial

volume is expected to be $79 \text{ GeV}/c$ with a mean beam momentum for K_L which decay of $46 \text{ GeV}/c$.

HIKE Phase 2 is planned to follow after Phase 1, which is scheduled after LS4 and to run up to LS5. The expected protons per spill in Phase 2 is six times that of NA62 at 2×10^{13} . Assuming 200 days of data taking and 3000 spills per day (as in Phase 1) and 3.2×10^{-6} K_L decays in the Fiducial Volume (FV) per proton on target would result in 3.8×10^{13} decays per year. The expected time to Phase 2 goals is 6 years of data taking.

0.1.3 Phase 3 (KLEVER)

HIKE phase 3, previously known as KLEVER [5], is an experiment designed to measure the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay. The experimental setup is shown in Figure 4 and differs from the design used at NA62 and HIKE Phase1/2. The setup consists of a large number of photon veto detectors located around a 160 m long vacuum volume to produce a complete coverage of photons produced from K_L decays. The detectors include coverage for photons emitted up to 100 mrad and allow for photons emitted with an angle up to 7.5 mrad to be detected by the main electromagnetic calorimeter (MEC).

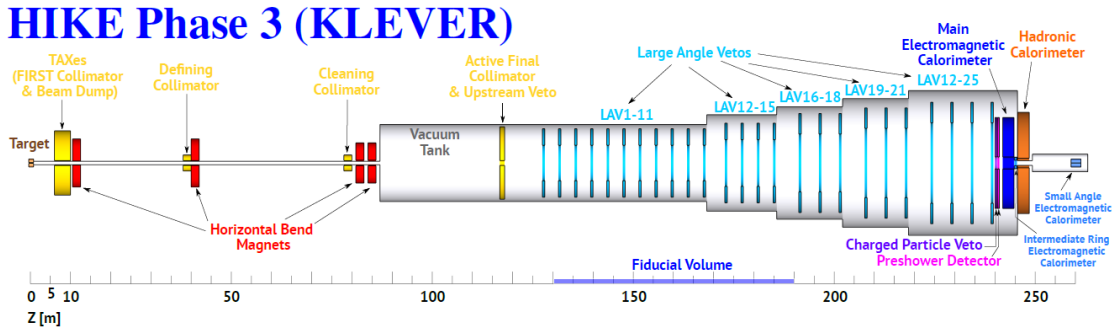


Figure 4: Diagram of HIKE phase 3 detector setup [2].

Due to the significantly different experimental setup, Phase 3 is separated from the HIKE proposal submitted in 2023 [2]. However, some detectors would be shared between the two experiments as the granularity and efficiency requirements for Phase 3 have driven the detector requirements for the first two phases, reducing the overall costs. The LAVs, MEC, IRC, SAC and the hadronic calorimeters, which are to be built for HIKE Phase 1/2, had all designs driven by their usage in Phase 3. The

upstream veto, charged particle veto, and the pre-shower detectors are unique to Phase 3.

0.2 Flexible Software

As briefly described in Section ??, the software for NA62 is split into three main C++ packages. NA62MC is a GEANT4-based Monte Carlo simulation package that simulates the NA62 experiment. The Monte Carlo (MC) utilises functions within GEANT4 to construct the sub-detectors of NA62, including all their materials, volumes and structures. It is possible to select a number of kaon decays to simulate in order to create signals similar to what are produced in reality at the experiment. After the simulation is complete, the software saves the signals in each detector in a ROOT file for use in reconstruction or analysis. NA62Reconstruction is then used to reconstruct either the Monte Carlo data or collected data from the experiment, producing hits within the detectors and clustering them into candidates. NA62Analysis is used to analyse the data produced from the reconstruction to conduct all the physics studies that are performed at NA62.

The NA62FW was designed for use only with the NA62 experimental setup, resulting in minimal configurability and flexibility. Within the simulation, the only option for users to modify was to enable/disable sub-detectors. With the experiment foreseen to run up to LS3, an upgraded software package is required to allow for the development of new detectors for future kaon experiments. The software had to be flexible and configurable enough to allow for the simultaneous use for NA62 data taking and physics analysis, as well as for the development of new experiments. To implement a flexible framework, all detector classes are refactored to inherit from a common class, which contained the essential functions and variables for all detectors. This allowed for all classes of the detectors to be parsed using the same machinery. The input to the simulation is changed from a simple text file, to a single configuration file for all the detectors. The configuration file contained any arguments required for individual detectors which are defined in Table 2, including detectors upgraded later on in this Chapter. The X, Y, and Z placement of the back face of the detectors, along with the rotation, are also defined in this file. An

example entry in the configuration file for a STRAW station is shown in Figure 5, where a name is given for the station "STRAW0", along with a chamber ID, chamber type defining the size of the straws, and the chamber geometry which determines if the beam hole is symmetric or not.

```

Name           Arguments      X Y    Z      θx θy θz
Spectrometer("STRAW0", 0, 0, 0) 0 0 183705.9 0 0 0

```

Figure 5: Example configuration for the STRAW0 in the geometry file for NA62MC.

Detector Name	Arguments
STRAW	Chamber Name: "STRAW0", "STRAW1", "STRAW2" or "STRAW3" Chamber ID: 0, 1, 2 or 3 Chamber Type: 0 (10mm) or 1 (5mm) Chamber Mode: 0 (default) or 1 (symmetric)
Spectrometer Magnet	Magnetic field intensity
Vessel	Length InnerRadius OuterRadius Type: "BlueTube", "BlueTubeNoRails" or "GreyTube"
GigaTracker	PixelType: 0 (NA62), 1 (NA62/2), 2 (NA62/3)
RICH	Type: "NA62" or "HIKE" SiPMTType: "3mm", "6mm" or "9mm"

Table 2: Arguments for detectors in the geometry file for use within NA62MC.

When constructed inside the simulation, each detector is contained within its own separate outer volume called the "Responsibility Region" (RR). The creation of the RR is a simple task for most detectors apart from the STRAW and LAV which are made up of multiple stations that are not continuous, resulting in multiple responsibility regions for each station.

0.2.1 Testing: Geometry and Magnetic Fields

After the simulation was upgraded, all the geometries of each detector had to be validated to ensure that they are still correct with respect to the previous version. This is due to the complex nature of refactoring the detectors with a number of

volume changes and the way the detectors are constructed. The validation is completed using a virtual particle within the GEANT4 framework called a Geantino. The Geantino is a particle that traverses the simulation in a straight line and interacts with all volumes present in the simulation but will not be affected by magnetic fields. A large square beam of these particles is sent through the simulation, and every time the particle passed across a boundary between two volumes, its position is saved. Using this information it is possible to reconstruct the entire geometry of each detector in order to check the placement and geometry to mm precision to ensure that the simulation is still accurate. An example of the Geantino hits on a single view of STRAW chamber 1 is shown in Figure 6, where it is possible to see the features of the view, such as the missing straws within the centre used to produce the chamber beam hole. For the full validation, a larger sample of Geantinos is used to achieve the required precision. However, if there is a discrepancy between the new version and the previous, a smaller area is selected to create a more precise picture of what is incorrect. This information was then used to modify the software to fix any geometry issues. This check is conducted for all sub-detectors within the simulation.

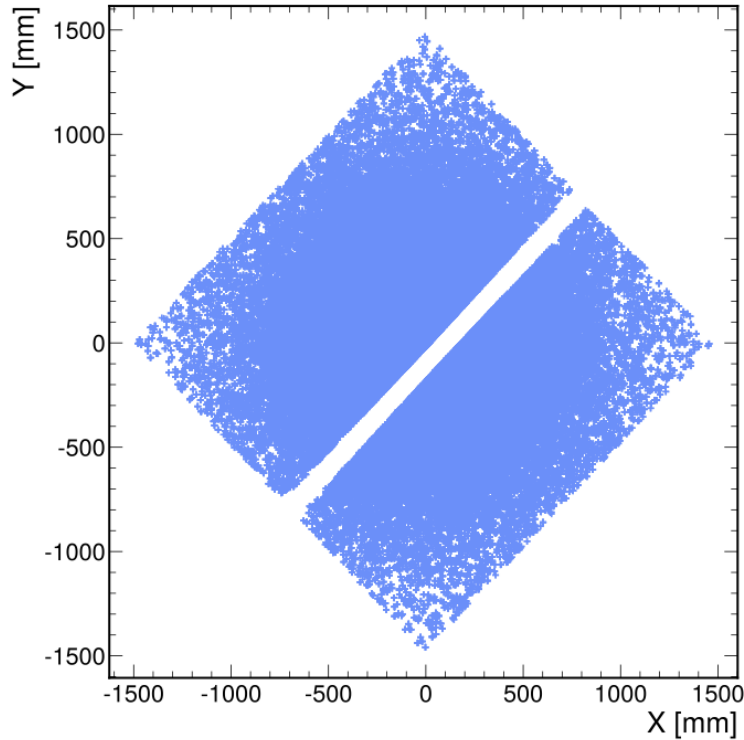


Figure 6: Geantino hits on a single view in STRAW chamber 1.

To reduce the chance of overlaps and to clean up the simulation, the size of the unused space in all sub-detectors surrounding the physical geometry is reduced. The RR for all sub-detectors is reduced to just slightly larger than the physical detector. It is possible to see this change in figure 7, which depicts the direction of the magnetic fields. As the magnetic fields are only calculated in volumes, it is possible to see the reduction in RR size around the geometry. In 7(top) the field is calculated outside of the actual physical sub-detector due to the larger RR, whereas in 7(bottom) the RR has been reduced to a few mm larger than the actual physical detector. As well as reducing the chance of overlaps, reducing the RR would speed up the simulation slightly due to the reduction in unnecessary calculations for the magnetic field outside of the detector geometry. To improve the usability for users a geometry check is completed every time before the simulation starts. Each outer detector volume is checked for any overlaps with other detectors in all three dimensions, taking into account any rotations and the shape of the outer volumes. If an overlap of two sub-detectors is found, the simulation does not start, and an error is produced to the user describing where the overlap is and what sub-detectors are involved.

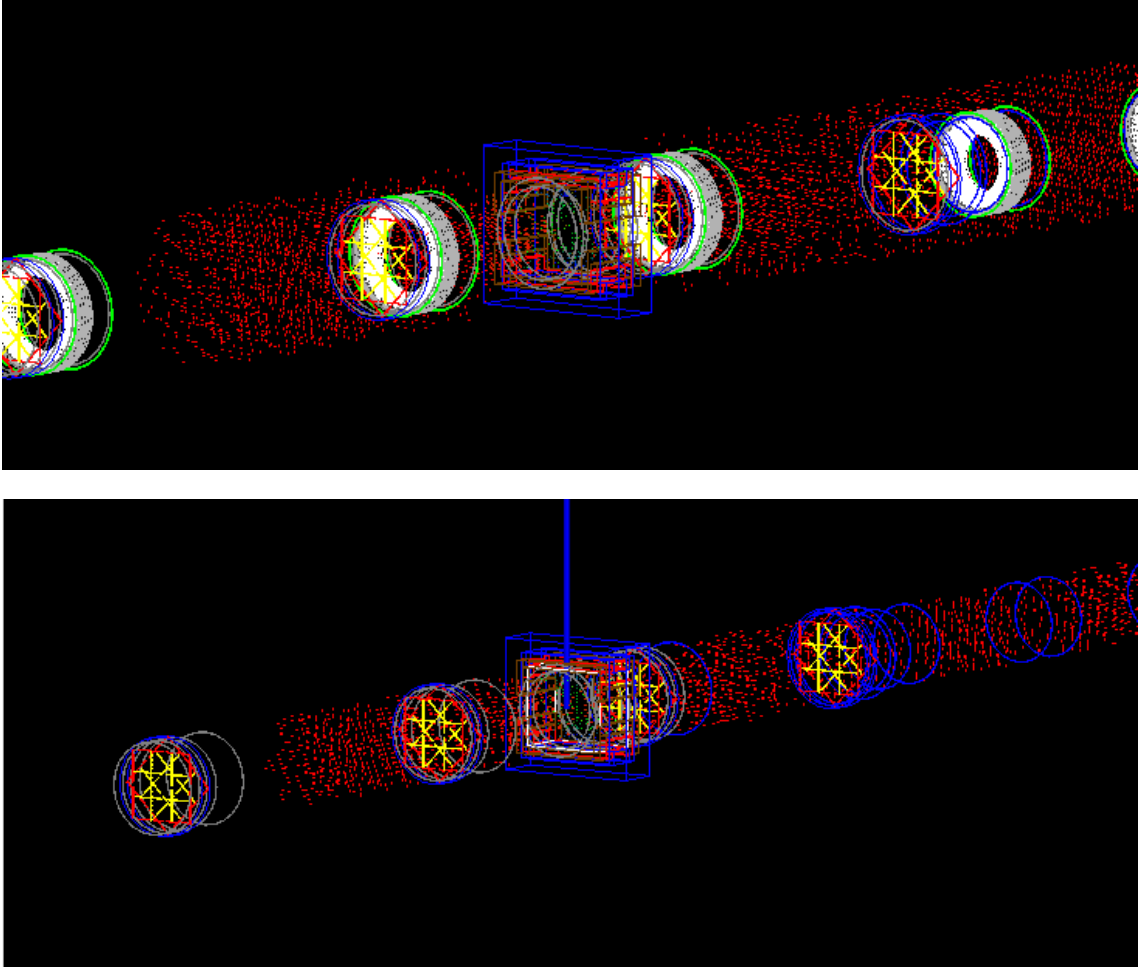


Figure 7: Visualisation of the magnetic fields in the spectrometer region from the simulation. Top, previous version with large outer volumes and bottom, new flexible version with reduced outer volumes. The red arrows represent the direction of the magnetic field in the volumes.

The magnetic field in the simulation is based on measurements of the magnetic field at NA62 from hall probes [6], with this information saved in the form of data files used as inputs to calculate the magnetic field at various points in the simulation. In NA62MC the magnetic field is split into three distinct regions: the Bluetube field within the decay region which is used for small corrections, the fringe field located at the edges of the MNP33 magnetic field and the MNP33 field which is the main magnetic field defined in the simulation from the MNP33 dipole magnet. In the previous version of the simulation, the position of the magnetic fields is written within the static code. However, with the flexible version, it is possible to move the MNP33 magnet to any position via the geometry input file. This required the magnetic field to be calculated relative to the central position of the MNP33 dipole

magnet. An example of this flexibility is shown in Figure 8 where the MNP33 magnet is moved from $Z=198\text{ m}$ to $Z=163\text{ m}$, which is in front of the STRAW chamber 1. The MNP33 and fringe magnetic fields is recalculated from the new position at 163 m to account for the placement of the MNP33 magnet.

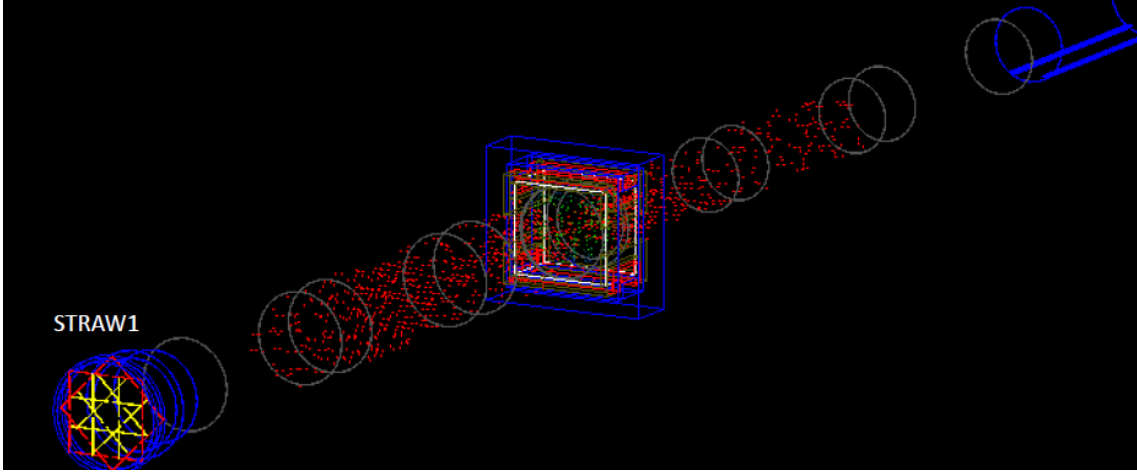


Figure 8: Visualisation of the MNP33 magnet placed in front of the STRAW chamber 1 at $Z=163\text{ m}$ from the target in the simulation. Originally, the magnet is placed at $Z=198\text{ m}$, which is after the STRAW chamber 2. The red arrows represent the direction of the magnetic field in the volumes.

Similar to the changes in the sub-detector construction, the modification of the magnetic fields required validation. This is done using similar machinery to the geometry test completed previously but with a different GEANT particle, the charged Geantino. This is similar to a normal Geantino, where the particle traverses all volumes within its path but interacts with magnetic fields. As the charged Geantino enters a magnetic field, it is deflected as if it is a positively charged particle. This test ensured that the same tracks of the charged Geantinos between the previous version and the new flexible version are being produced within the magnetic fields. An example of the tracks taken by the charged Geantinos within the simulation is shown in Figure 9, where it is possible to see the displacement of the particles within the magnetic field regions. The exact XY positions of the hits would then be validated against a sample of charged Geantinos with the exact same characteristics and starting parameters produced with the previous version of the software.

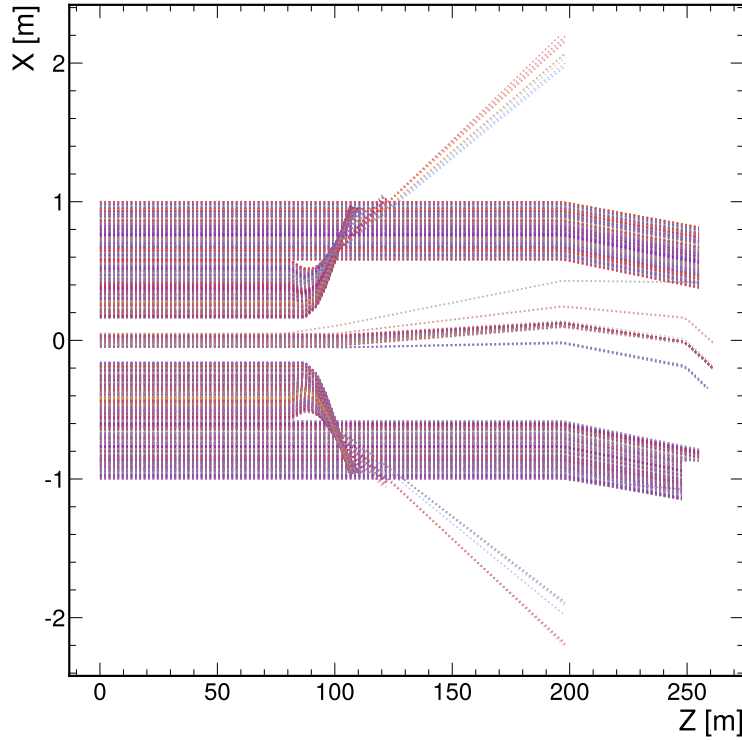


Figure 9: Tracks for charged Geantinos with a $400 \text{ GeV}/c$ momentum through the NA62 detector.

With minimal modification it would also be possible to add the functionality to add multiple magnets within the simulation, producing a combined field. After the geometry and magnetic field tests are completed, full test productions of $K^+ \rightarrow \pi^+\pi^0$, $K^+ \rightarrow \pi^+\pi^-\pi^+$ and $K^+ \rightarrow \mu^+\bar{\nu}$, are produced to test the validity of the existing NA62 setup along with the entire software chain (MC→Reconstruction→Analysis).

After testing, the software is ready to be used to develop new detectors and produce physics studies for HIKE.

0.3 The KTAG upgrades for HIKE

To sustain the increased kaon rate, the KTAG required upgrades to the photon-detection and data acquisition systems, allowing to reach the required time resolution of 15-20 ps and kaon identification efficiency of greater than 95 %. The expected kaon rate in HIKE Phase 1 is 200 MHz, which corresponds to a maximum rate of

10 MHz/cm² for the detected photons at the PMT array. This rate takes into account the production of Cherenkov photons in CEDAR-H and the resultant acceptance of the photons on the PMT plane. It is proposed that the NA62 PMT arrays are replaced with arrays of Micro-Channel Plate Photomultipliers (MCP-PMTs). The MCP-PMTs would provide a single photon time resolution of 50-70 ps depending on the model, a low dark rate of 1 kHz/cm² and the ability to sustain the required photon rate. A downside of the MCP-PMTs is their limited lifetime from the ageing of the photocathode caused by feedback ions from the residual gas resulting in a rapid reduction in the Quantum Efficiency (QE) [7]. This is measured as degradation of PMT QE with the increasing Integrated Anode Charge (IAC). MCP-PMTs are similar to normal PMTs except that the dynode section is replaced by a micro-channel plate. An example diagram is shown in Figure 10, where a photon hits the photocathode window, producing a single electron, which is then accelerated towards the MCP by an electric field provided by a voltage difference across the gap. The electron then collides with the walls of the MCP, producing secondary electrons, which results in a cascade collected by the anodes below. The dual layer design, or chevron, adds additional benefits compared to a single layer design such as better gains due to the increased surface for secondary electrons to be produced.

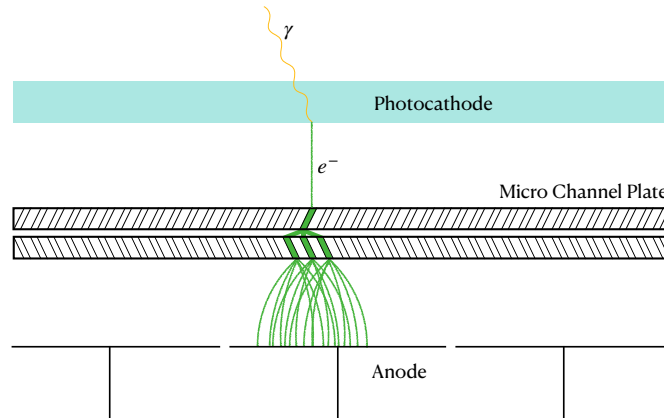


Figure 10: Diagram describing the workings of a MCP-PMT.

Major breakthroughs to improve the lifetime of the MCP-PMT have come with the method of Atomic Layer Deposition (ALD) [8], where the channels of the MCP are coated with a ultra-thin layer of Al₂O₃ or MgO. The layer prevents the outgassing from the MCP glass substrate and drastically reduces the heavy ions reaching

the photocathode resulting in significant increases in lifetime of the MCP-PMTs [9]. Utilising the method of ALD, MCP-PMTs from Photonis have been demonstrated to withstand an IAC greater than 20 C/cm^2 [10], corresponding to a lifetime improvement of a factor of 100 compared to standard MCP-PMTs, shown in Figure 11.

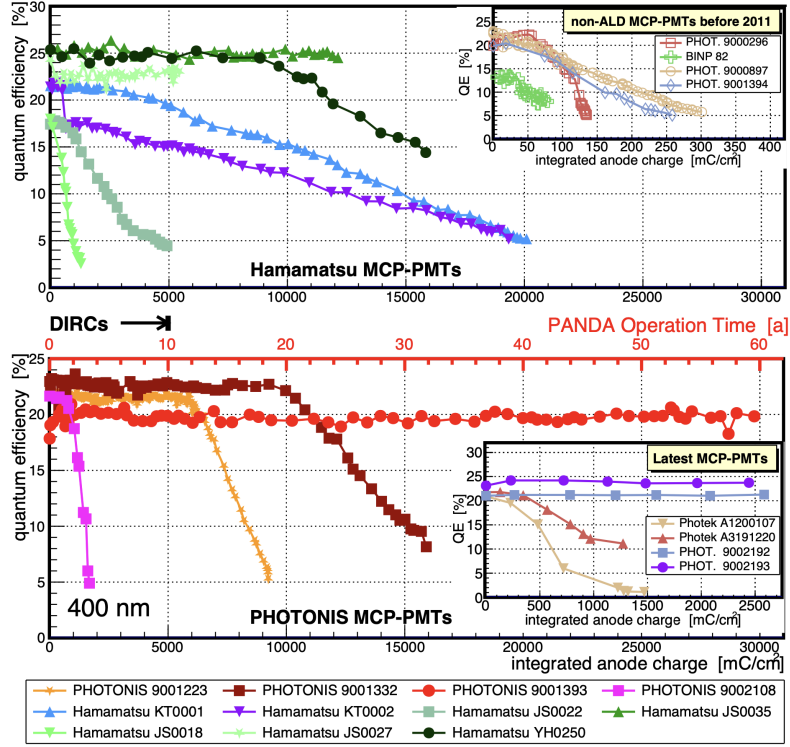


Figure 11: Lifetimes of available MCP-PMTs with the photocathode QE vs IAC for photons at 400 nm wavelength. Top, old ALD coated MCP-PMTs from Hamamatsu with non coated versions shown in the insert plot. Bottom, more recent ALD coated MCP-PMTs from PHOTONIS and Photek in the inserted plot. [11]

An example of MCP-PMT model proposed for the KTAG is shown in Figure 12. The module consists of an 8×8 array of pixels which can individually be read out.

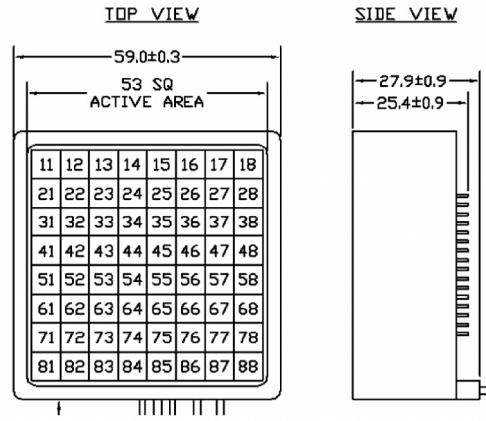


Figure 12: Schematic of Photonis MCP-PMT [12].

Simulations of the KTAG with the upgraded photon-detection system for use in HIKE Phase 1 show that each of the eight KTAG sectors would use four of these MCP-PMT, shown in Figure 13. The four MCP-PMTs per sector are required for full coverage of the light spot produced when the light exiting the CEDAR is reflected by the spherical mirror. The schematic in Figure 12 shows there is some unused space around the active area of each MCP-PMT, which would result in reduced efficiency, especially in the centre, where the maximum amount of light would be expected. To mitigate this, reflective surfaces are attached to each side of the individual MCP-PMTs shown in Figure 13 to reflect the light onto the active surface.



Figure 13: Visualisation of the MCP-PMT array placed inside of the KTAG

The resultant light spot for one of the sectors is shown in Figure 14, where the inactive areas between the individual MCP-PMT and the increased amount of light at the edges produced from the reflective surfaces are shown. The position of the MCP-PMT in the KTAG sector is optimised to produce the maximum number of photo-electrons per kaon candidate.

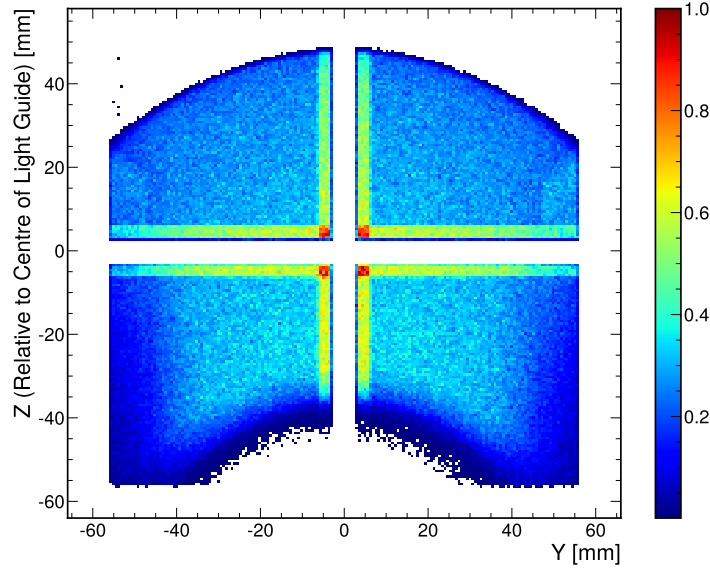


Figure 14: Collected light distribution for the MCP-PMT installed in the KTAG.

HIKE Phase 1 required the MCP-PMTs installed within the KTAG photon-detection system to have a demonstrated lifetime of 50 C/cm^2 . Other requirements set by HIKE for the MCP-PMT is a single photon detection rate of 10 MHz/cm^2 without degradation to the signal due to charge saturation effects. The single photon detection requires a high amplification factor (gain) of $\mathcal{O}(10^6)$. The MCP-PMTs manufactured by Photonis with 2 layers of ALD provide the best lifetime with no degradation in QE after 30 C/cm^2 , shown in Figure 11. An excellent candidate for MCP-PMT is Photonis XP85012-S-R2D2 of the PLANACON line. However, the Photonis MCP-PMT doesn't match with the HIKE Phase 1 rate requirements and result in reduced gain above 1 MHz/cm^2 , shown in Figure 15. It is possible to reach 10 MHz/cm^2 with smaller sensor size but would result in a quadrupling of the cost to cover the light spot. Future research looking at reducing the resistance, capacitance and gain for single photon detection could possibly allow the XP85012-S-R2D2 to accommodate up to 10 MHz/cm^2 .

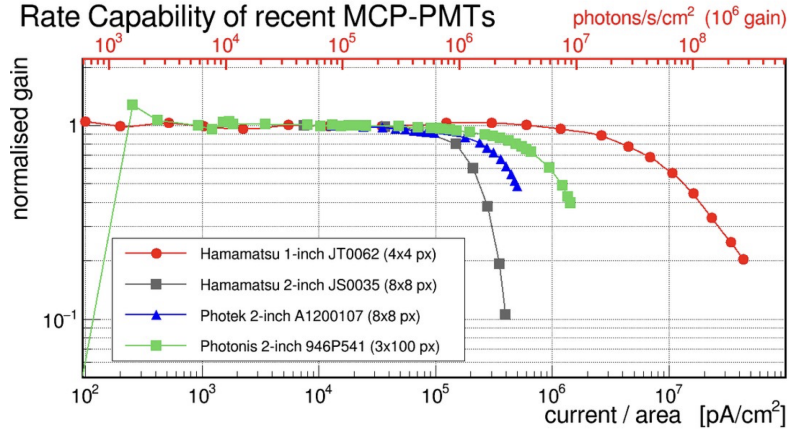


Figure 15: Rate capability of state of the art MCP-PMTs [13].

The Photonis MCP-PMT has a Collection Efficiency (CE) of above 60 %, the probability of the photo-electron passing through the pores of the MCP and producing a signal. With current technologies it is possible for MCP-PMTs to reach CE of over to 90 % [14], however the Photonis MCP-PMT is selected as it better fits the HIKE Phase 1 timing requirements as the higher CE models utilise late photo-electrons resulting in higher time resolutions. The XP85012-S-R2D2 MCP-PMT has a transit time spread of 50 ps with a timing precision RMS of 120 ps [15] and results in a time response, shown in Figure 16 which is implemented for the HIKE simulations.

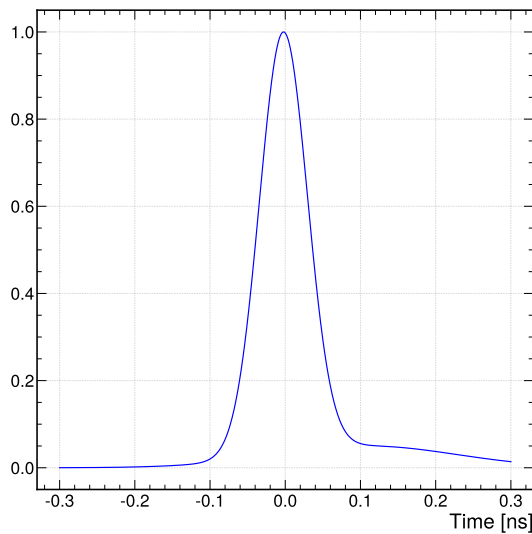


Figure 16: Time response of the Photonics XP85012-S-R2D2 MCP-PMT.

The QE for the MCP-PMT from Photonis is shown in Figure 17. There are

two options for the photocathode of the MCP-PMT available: a standard Bialkali (Sb-Rb-Cs, Sb-K-Cs) or a Hi-QE Blue. The HI-QE are more recent models, however there is no lifetime data available so albeit being the preferred option the standard Bialkali is chosen for use in the simulation. Further studies at the University of Birmingham are planned to characterise the Photonis XP85012-S-R2D2 MCP-PMT by measuring the lifetime, QE, CE, and other characteristics in order to decide which type would be used for high-intensity kaon physics at HIKE.

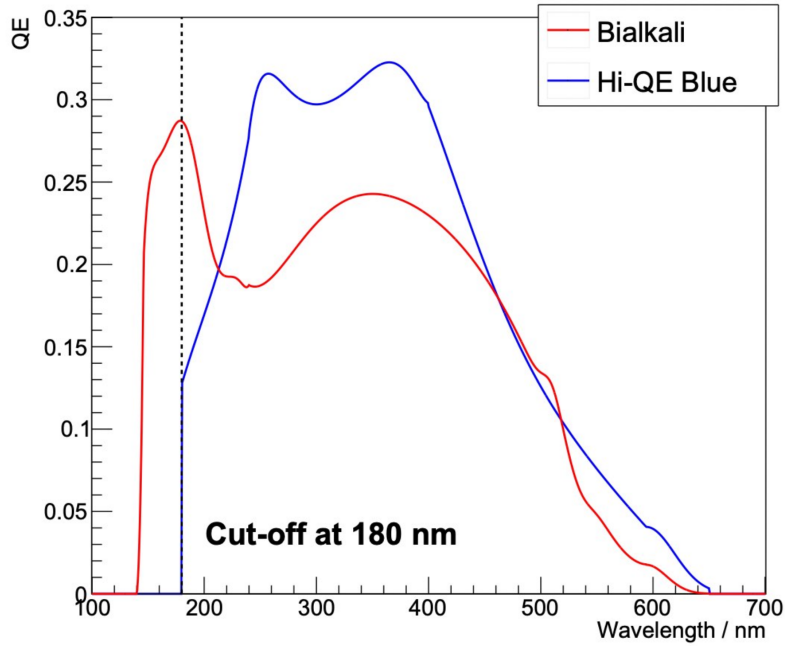


Figure 17: Quantum Efficiency of Photonics MCP-PMT as a function of wavelength.

From the Monte Carlo simulation, the implemented MCP-PMTs performed excellently with the result shown in Figure 18. The MCP-PMT produced 24.3 photoelectrons per kaon candidate with a time resolution of 18 ps, taken from the Gaussian sigma of the distribution in Figure 18 (Right). The results surpassed the requirements for the KTAG in HIKE and the upgrades are ready to be used in the HIKE simulations providing the lifetime and rate capability are acceptable.

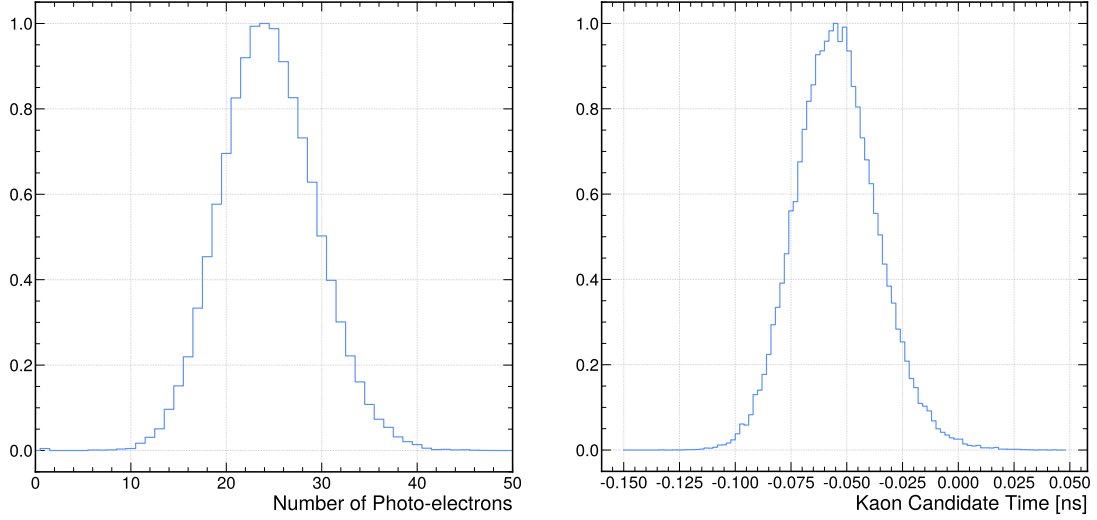


Figure 18: Results from the Monte Carlo simulation of CEDAR-H and MCP-PMT. Left, the number of photo-electrons per kaon candidate and right, the Kaon Candidate time.

0.4 RICH Upgrade

The overall design of the NA62 RICH described in section ?? is well suited for re-use within HIKE with an upgrade to the photon-detection system. The NA62 RICH photon detection system comprises two arrays on either side of the beam pipe, equipped with Hamamatsu R7400-U03 PMTs placed in a pattern described in Figure 19. The PMTs have a single hit time resolution of 240 ps and a quantum efficiency of 20 %. Due to the placement of the PMTs, the distance between the centre of each PMT is 18 mm, which is constrained by the sensor size and the geometry of the PMT itself. This distance between sensors is the main contributor to the single hit position resolution of 4.7 mm, resulting in an overall ring radius resolution of 1.5 mm. The number of hits per ring candidate and the single hit time resolution produced a time resolution of 80 ps. Due to the increased beam intensity, the main requirement of the RICH for HIKE is a time resolution of 20 - 30 ps.

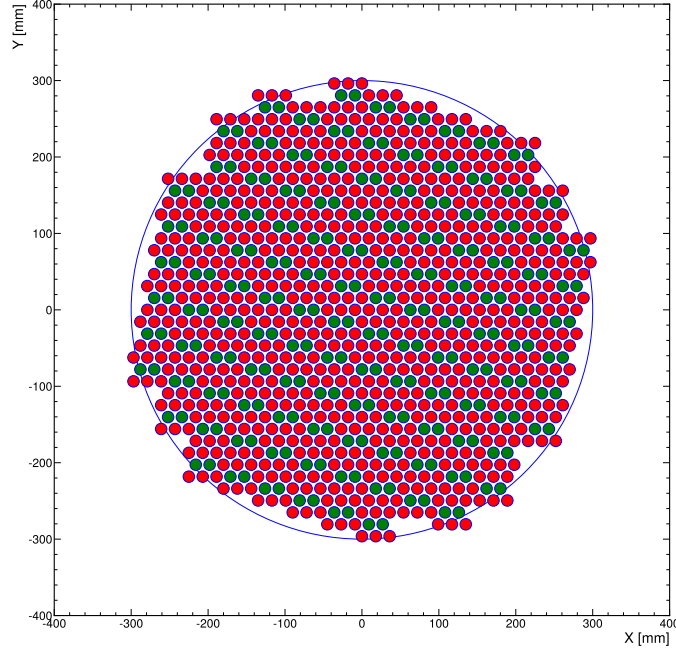


Figure 19: Diagram of RICH PMT array

The proposed RICH photon detection system upgrade would replace the current PMTs with a smaller sensor size, allowing for a smaller distance between sensor centres and reducing the hit position resolution. An excellent candidate for a sensor replacement are silicon photomultipliers (SiPMs). SiPMs with small sensor sizes, high quantum efficiency of 40 % and a single hit time resolution of at least 100 ps are available from major manufacturers. Three sensor sizes are selected for further investigation, $9 \times 9 \text{ mm}^2$, $6 \times 6 \text{ mm}^2$, and $3 \times 3 \text{ mm}^2$. Each SiPM sensor size had apparent benefits and downsides, and a balance between performance and complexity/price had to be taken into account. The smaller sensors would produce a better hit position resolution but at an increased cost and complexity. Whereas the larger sensors have worse hit position resolution but at reduced complexity and cost. A study is conducted to determine the optimal sensor size for the SiPMs, balancing overall detector performance with cost and complexity.

Using the upgraded NA62 software package described in Section 0.2, it is possible to implement the new PMTs into the RICH Monte Carlo simulation with the implemented arrays shown in Figure 20. Compared to the PMT arrays currently

used in NA62, the PMs are aligned on the vertical axis to simplify the placement, allowed by the square geometry of the new PMTs. The area of the expected light spot, shown in Figure 20, as a grey circle, is completely covered by the SiPMs with no gaps between the sensors which is an improvement compared to the NA62 setup shown in Figure 19.

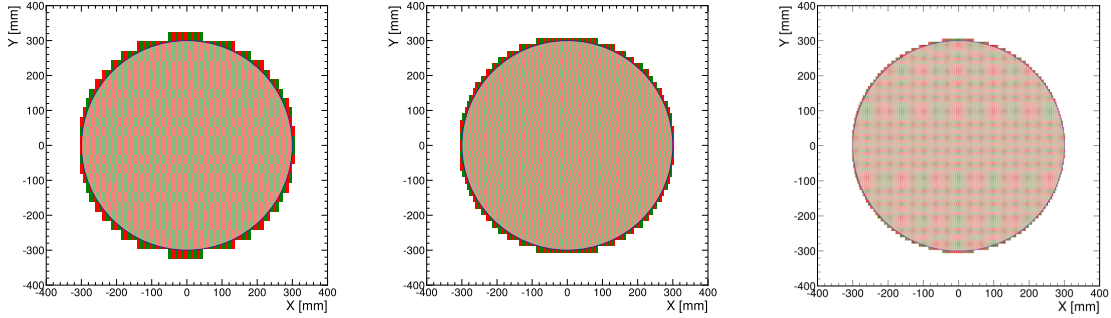


Figure 20: Diagram of proposed RICH SiPM arrays for use in HIKE. Left, 9x9 mm, center 6x6 mm and right 3x3 mm SiPMs.

The performance for HIKE required a SiPM with a Quantum Efficiency (QE) of at least 40 % [2]. The Hamamatsu S13360-**50CS series is selected as the preliminary candidate for the RICH PMT upgrade to achieve this requirement. Hamamatsu offers a number of SiPMs in the S13360 series, with the **50CS selected for its improved Ultra Violet (UV) sensitivity and high quantum efficiency of 40 % at 460 nm, shown in Figure 21.

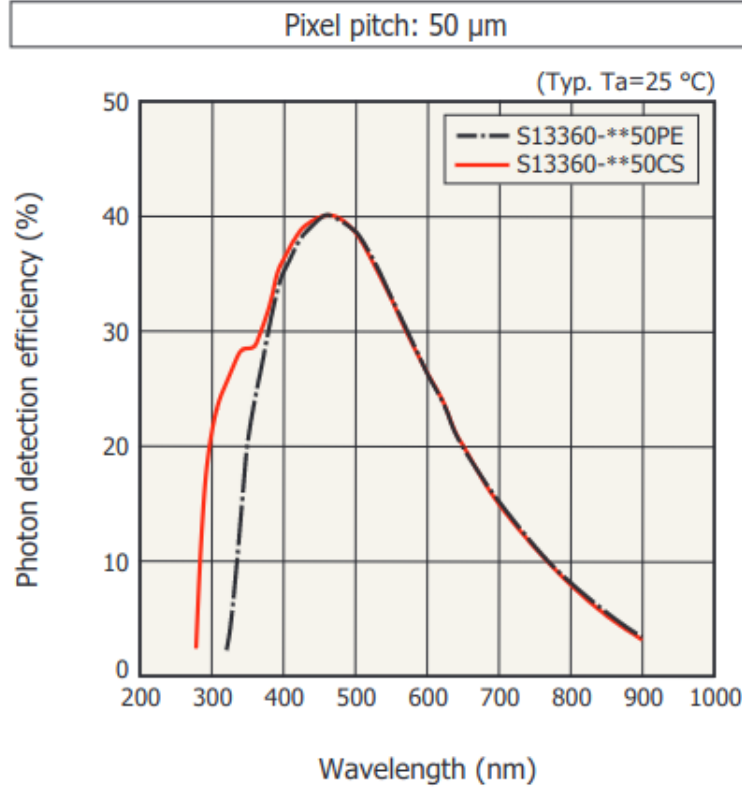


Figure 21: Selected SiPM quantum efficiency taken from Hamamatsu S13360 data-sheet [16]

Along with the QE efficiency, a single hit time resolution of 100 ps is required [2]. The manufacturer stated the Hamamatsu S13360-**50CS series has excellent time resolution, but no definite value is given. The time resolution also depends on the final design of the RICH readout electronics and would have to be measured in a test setup. The SiPMs from Hamamatsu have previously been tested and have reached a single hit time resolution below 100 ps [17]. For use in the simulation a conservative simple Gaussian time response curve with a sigma of 100 ps is used, which would be updated later on for a specific model to be implemented when chosen.

The expected performance of the RICH detector with the new SiPMs is compared to the current NA62 RICH detector in Table 3. With the increased QE, the expected number hits per π^+ between 15 GeV/ c and 45 GeV/ c was expected to increase by a factor of two. This, along with the improved time resolution of the new SiPMs resulted in an expected π^+ time resolution of 27 ps at 15 GeV/ c and 20 ps at 45 GeV/ c . Using the implemented upgrades to the RICH detectors, the above expectations are confirmed by a Monte Carlo sample of $K^+ \rightarrow \pi^0 e^+ \nu$ (K_{e3}).

	NA62 RICH	HIKE RICH
Sensor time resolution	240ps	100ps
Sensor quantum efficiency	20%	40%
Number of photo-electrons for π^+ at 15 GeV/c	7	14
Number of photo-electrons for π^+ at 45 GeV/c	12	24
Time resolution for π^+ at 15 GeV/c	90ps	27ps
Time resolution for π^+ at 45 GeV/c	70ps	20ps

Table 3: Comparisons of the expected performance of the NA62 and HIKE RICH detectors [2]

For the three sensor sizes, a test production of 1M K_{e3} is used to determine the preliminary performance of the detector and compare this to the current NA62 setup. The plots in Figure 23 show the light distribution on the arrays for the three sensor sizes: 3 mm 23 (a,b), 6 mm 23 (c,d), and 9 mm 23 (e,f) compared to the illumination on the current NA62 arrays 22. Due to the better placement of the square sensors, it is possible to see a more uniform illumination of the PMT arrays and better coverage of the light ring, especially in the Jura side. It is clear to see that either sensor size would be an improvement on the current NA62 setup.

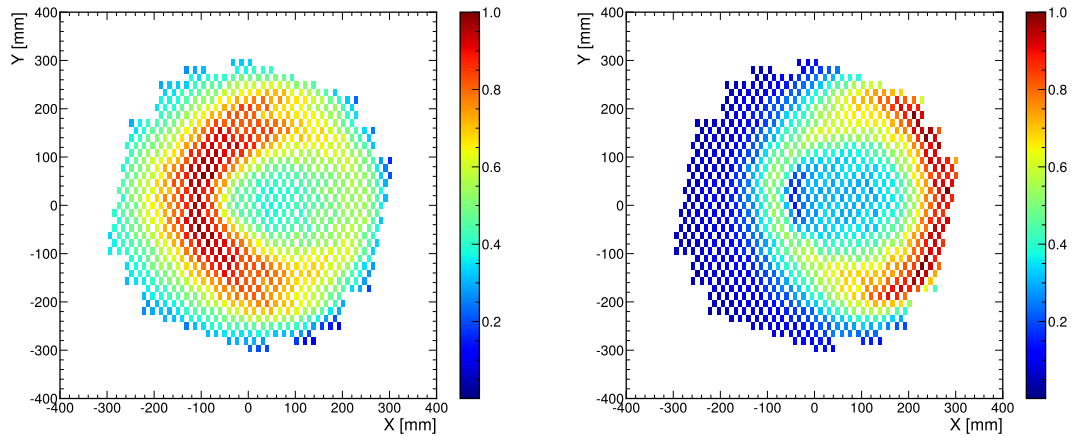
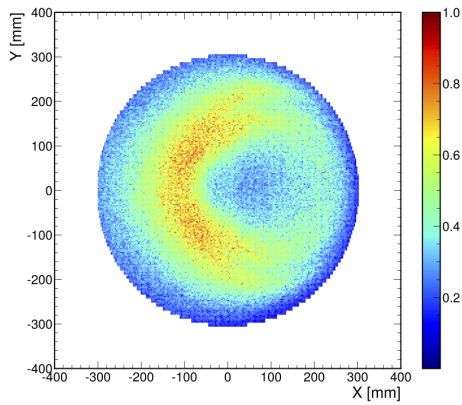
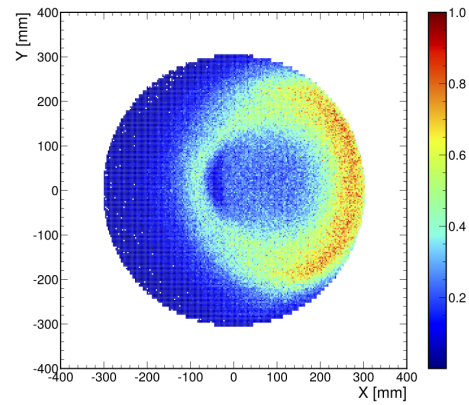


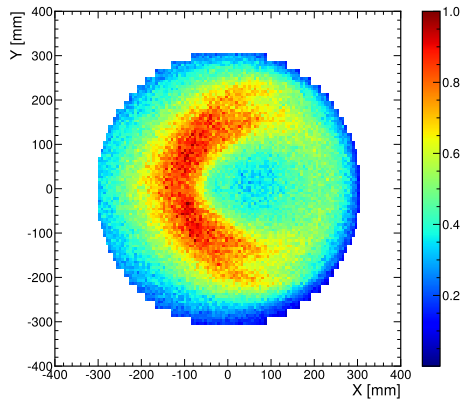
Figure 22: Cherenkov light illumination of RICH PMT arrays for the NA62 SiPM arrays produced from a MC sample of K_{e3} . The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.



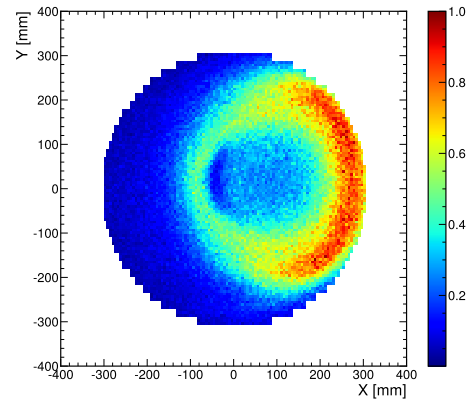
(a) 3 mm SiPMs Saleve.



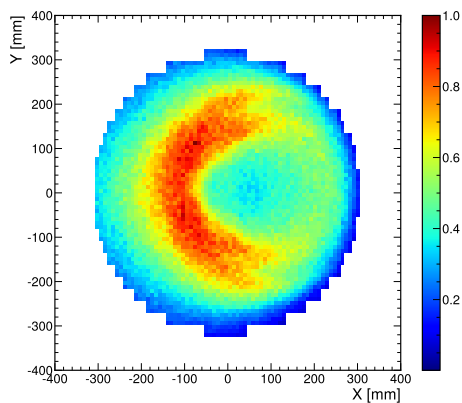
(b) 3 mm SiPMs Jura.



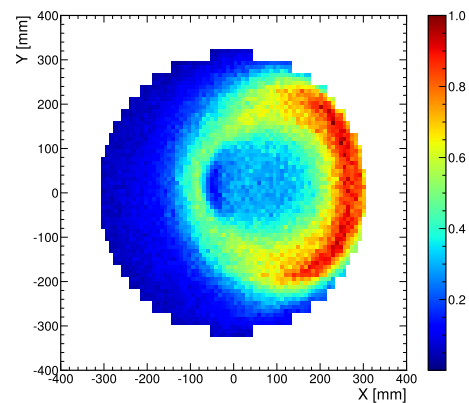
(c) 6 mm SiPMs Saleve.



(d) 6 mm SiPMs Jura.



(e) 9 mm SiPMs Saleve.

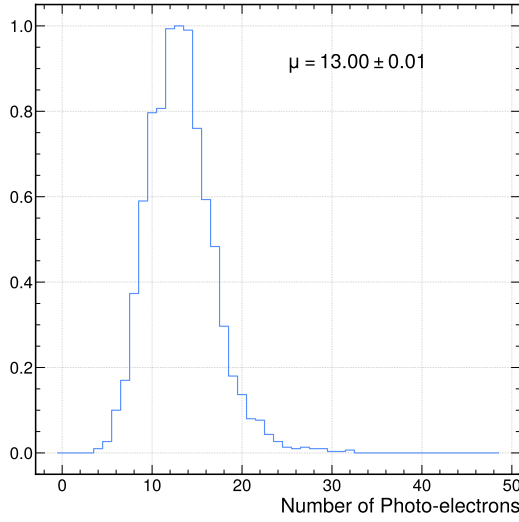


(f) 9 mm SiPMs Jura.

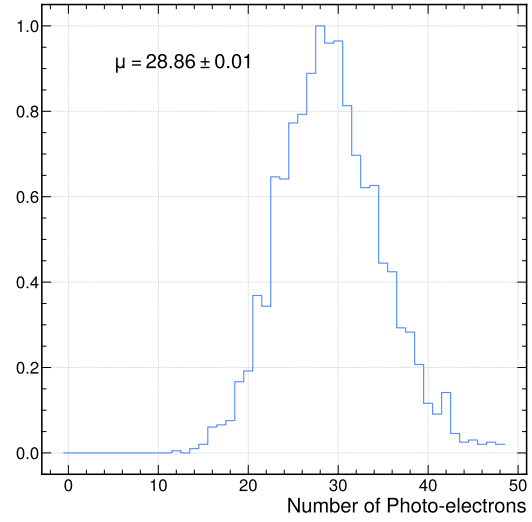
Figure 23: Cherenkov light illumination of RICH SiPM arrays for the three different sensor sizes produced from a MC sample of K_{e3} . The left column shows illumination on the Saleve side of the RICH detector, and the right column shows the Jura side.

In the reconstruction, the photo-electrons are clustered into ring candidates. The

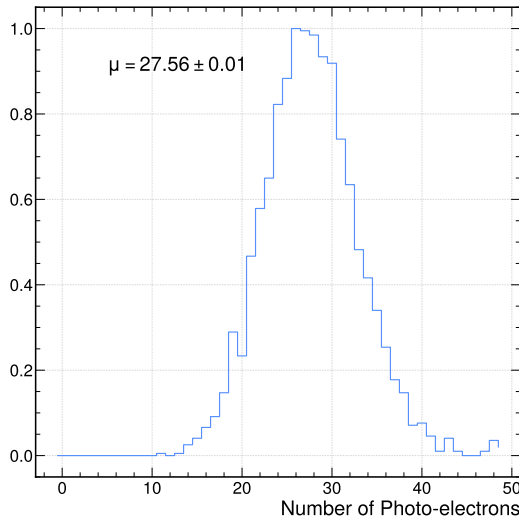
number of reconstructed photo-electrons is shown in Figure 24, with the SiPM sensors resulting in a two-times increase in the number of hits per ring candidate. The improved number of reconstructed photo-electrons is due to the enhanced quantum efficiency of the SiPMs and the sensors' better placement. This reduced the probability of multiple photons hitting the same sensor. There are minimal differences between the three sensor sizes shown in Figure 24 (b-d), with the 3 mm sensors having a slightly higher number of hits per ring candidate when compared to the 9 mm sensors.



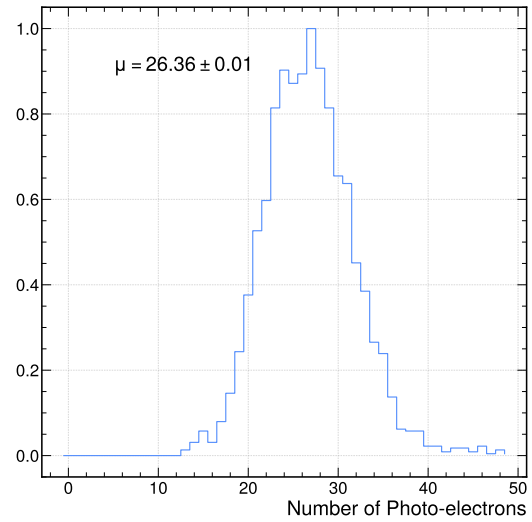
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.

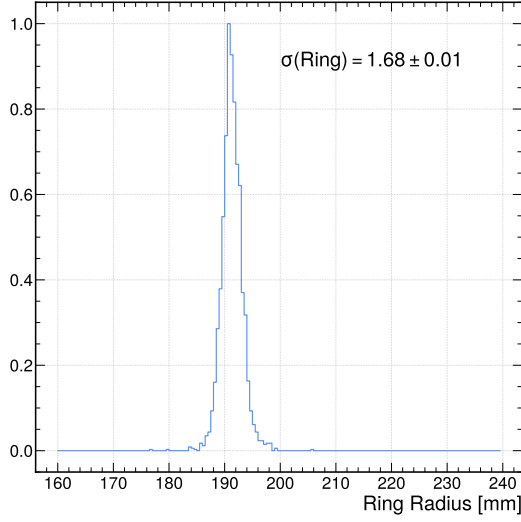


(d) 9 mm SiPMs.

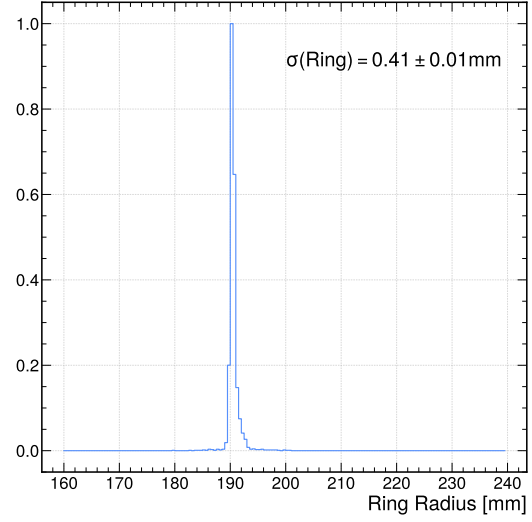
Figure 24: Number of photo-electrons per ring candidate for NA62 PMTs and the three possible HIKE SiPMs produced from a MC sample of K_{e3} .

The subsequent reconstructed ring candidate radius is shown in Figure 25, with the NA62 PMT array producing a resolution of 1.68 mm compared to 0.72 mm for the HIKE array with 9 mm SiPMs. With the smallest sensor size at 3 mm improving further to 0.41 mm. However, it would not be possible to obtain this resolution in reality due to small misalignments in the mirror and dispersion in the radiator gas. It is accepted that the 9 mm sensor size would provide the best possible resolution

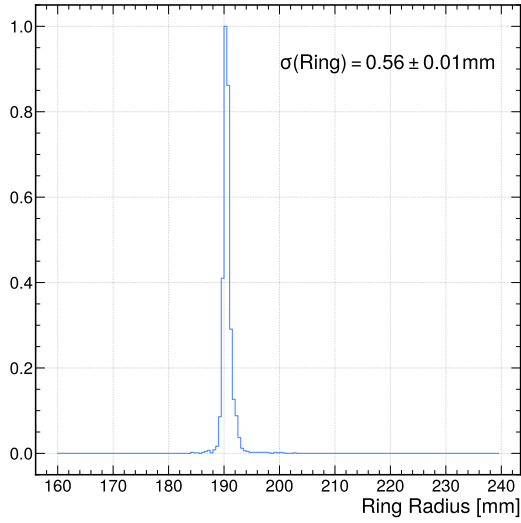
obtainable.



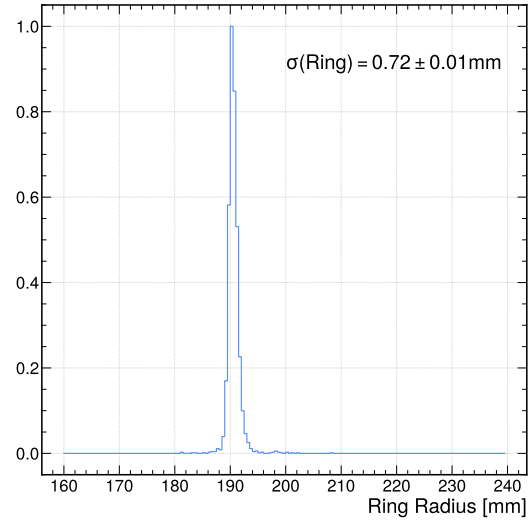
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.

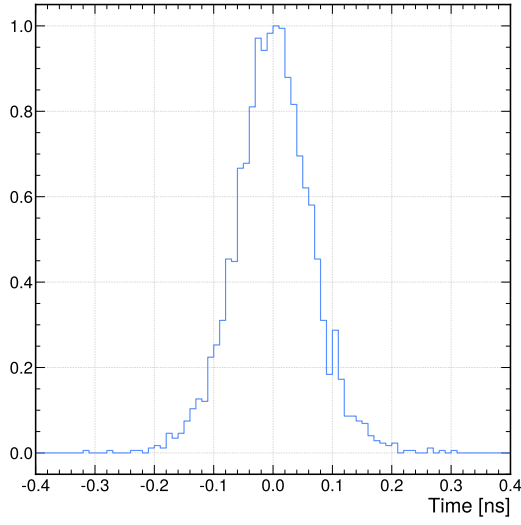


(d) 9 mm SiPMs.

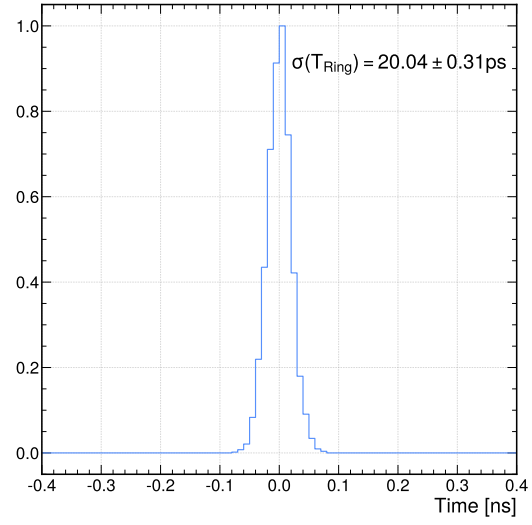
Figure 25: Reconstructed ring radius for electron candidates for NA62 PMTs and the three possible HIKE SiPMs produced from a MC sample of $K^+ \rightarrow \pi^0 e^+ \nu$ ($\text{Ke}3$).

The time resolution of the RICH detector is calculated using the K_{e3} sample by equally splitting the reconstructed photo-electrons of the ring candidates into two sets and calculating the time difference between them, with the results are shown in Figure 26. This method is used to remove any dependence on the reference time with respect to the single hit times. The 3x3 mm sensors produced the best time

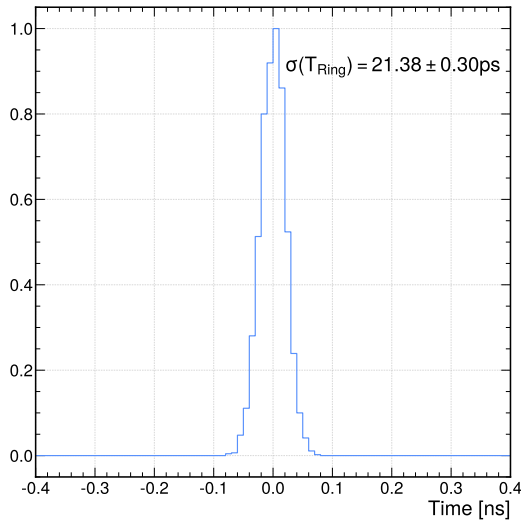
resolution of 20.04 ps compared to the 9x9 mm sensors with a time resolution of 21.22 ps. However, there is little difference between the SiPMs, with each sensor size producing a time resolution of less than 30 ps, fulfilling the requirements of HIKE. The improvement is clear with using the SiPMs when compared with the NA62 PMTs in Figure 26, with a time resolution of 61.35 ps.



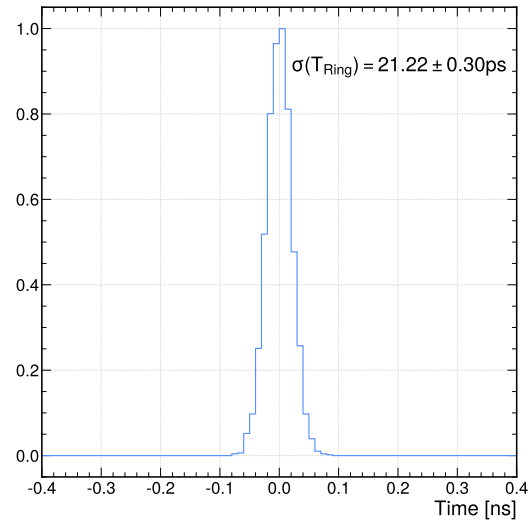
(a) NA62 PMT arrays.



(b) 3 mm SiPMs.



(c) 6 mm SiPMs.



(d) 9 mm SiPMs.

Figure 26: Time resolution for NA62 PMTs and the three possible HIKE SiPMs, calculated from the time difference between two equally populated subsets of hits within the ring candidate. A detailed description of the procedure is provided here [18]

After the RICH MC study, it is proven that introducing SiPMs would dramatically increase the granularity of the detector. All three SiPM sensor sizes produced results which exceeded the requirements for HIKE with a time resolution of around 20 ps and improvements on the reconstructed ring radius resolution. From the re-

sults of the HIKE simulation, the 9x9 mm SiPMs would be used as they provided excellent performance and would reduce the cost and complexity of implementing the smaller sensor sizes in the detector when it is constructed. A clear benefit is reducing the number of channels from 72,780 for the 3x3 mm² SiPMs compared to 8,588 for the 9x9 mm² SiPMs.

0.5 Detector Rates

After the new GigaTracker (GTK) and Ring Imaging Cherenkov counter (RICH) detectors were implemented into the simulation, it is necessary to determine the expected particle rate in the upgraded and non-upgraded detectors. The upgraded STRAW detector utilising 5 mm diameter straws is also included in the HIKE simulations. It is possible to estimate the expected rate for Phase 1 by scaling the known NA62 rates by the expected increase in beam intensity by a factor of 4 compared to NA62. The rates for Phase 2 are calculated using the new simulation software described in the previous sections.

An array of input files are produced in BDSIM [19] by the CERN beam group which contains the simulated HIKE beam components at a Z of 120 m, which is the start of the fiducial volume. The files are used as an input to the NA62MC software as the primary beam instead of the turtle beam, which is the default for NA62. Each file contains 10000 events, corresponding to 10000 protons on target. As there are 18K files produced in total, this is equivalent to 1.8×10^8 protons on target. As the statistics was not enough to obtain a reasonable estimate of the rates, each file is used 11 times as an input to the simulation with random seeds to increase the sample size. The simulation is run with the unforced flag allowing for four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$. After the simulation, the files are reconstructed using the NA62 Reconstruction software, and the rates are calculated by scaling the numbers in the output by the expected HIKE Phase 2 beam conditions, i.e, a spill time of 4.8 seconds and a beam intensity of 2×10^{13} protons per spill.

The K_L rate at the entrance of the fiducial volume is calculated to be 304.1 MHz. The rate of the four main K_L decays inside the fiducial volume is shown in Table 4.

KL decay	Rate [MHz]
$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0)	13.89
$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$)	9.63
$K_L \rightarrow \pi^0 \pi^0 \pi^0$	6.60
$K_L \rightarrow \pi^+ \pi^- \pi^0$	4.45

Table 4: Rates of the four main K_L decays in the HIKE phase 2 fiducial volume.

The total particle rates from K_L decays in the FV are shown in Table 5. To calculate the rate, all hits in each detector channel are collected and converted to the expected rate. The rates are large, but this is to be expected with a high intensity. However, the rate in the spectrometer for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay is concerning as a much lower rate is expected compared to the other decays due to the lack of charged daughter particles. Even with a lower branching ratio, it is expected that the $K_L \rightarrow \pi^+ \pi^- \pi^0$ decay should have a higher rate in the spectrometer due to the charged pions in the final state.

Detector	K_L Decay [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	126.8	79.9	50.5	49.2
STRAW-2	162.8	85.1	80.7	58.0
STRAW-3	165.7	89.3	67.7	56.1
STRAW-4	189.2	93.6	87.5	64.4
NCHOD	10.3	5.2	4.9	3.7
CHOD	90.5	33.4	61.3	33.8
IRC	3.8	1.2	4.2	2.0
LKr	609.5	266.0	440.7	298.3
MUV1	129.9	102.4	0.6	64.1
MUV2	10.3	17.9	0.1	4.7
MUV3	0.9	5.2	0.2	0.6
HAC	10.0	11.5	0.3	6.1

Table 5: Total rates in the detectors for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$.

Directly looking at the MC, it is possible to find the origin of the particles that produced the hits in the spectrometer. The production Z compared against the production radius of particles which produced a hit in the Spectrometer is shown in Figure 27. It is clear that for the $K_L \rightarrow \pi^0 \pi^0 \pi^0$ decay, the majority (80%) of the particles are produced after Z=210 m with most produced in the walls of the vacuum decay tube and the spectrometer chambers themselves, with only 20 % of particles produced before Z=210 m which would be consistent with being produced in the decay.

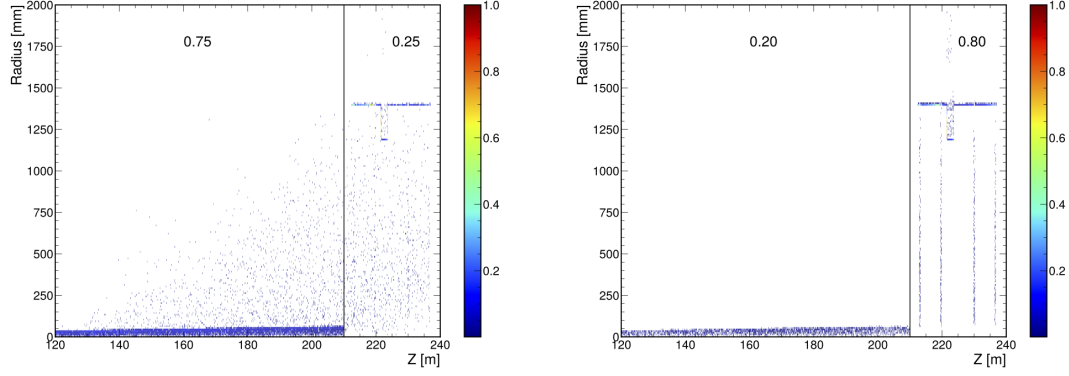


Figure 27: Production Z compared against production radius of particles producing a hit in the Spectrometer with the fraction of particles produced before and after $Z=210$ m. Left, for the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay and right, for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay.

This is also depicted in Figure 28 where the MC particle identification code of the particles producing hits in the spectrometer is shown. For the $K_L \rightarrow \pi^0\pi^0\pi^0$ the majority of the particles are electron/positron pairs, produced in interactions of photons in the walls of the vacuum decay tube and in the spectrometer chambers, compared to the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay where the majority of the particles are pions produced in the decay.

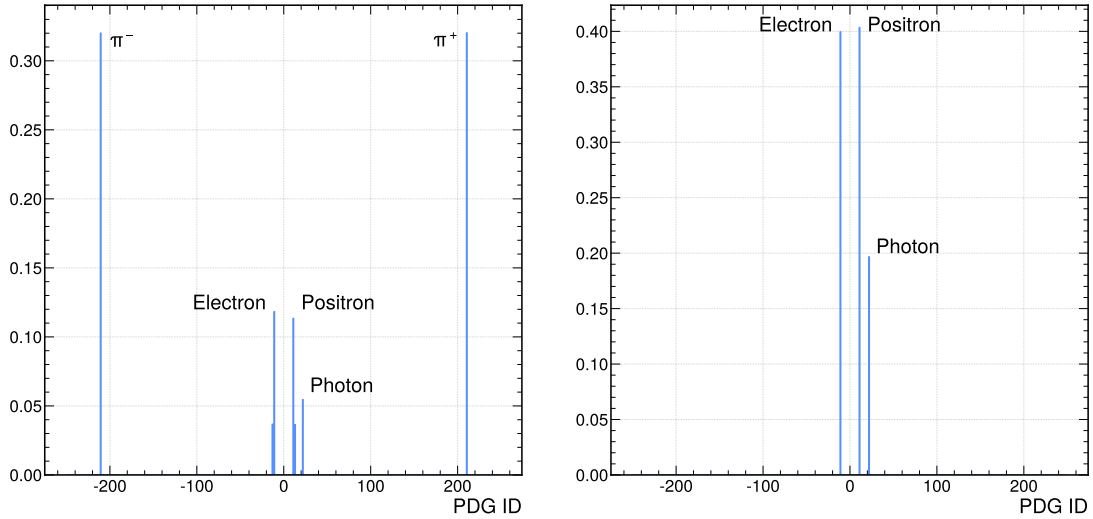


Figure 28: Monte Carlo particle code for the particles producing a hit in the Spectrometer. Left, for the $K_L \rightarrow \pi^+\pi^-\pi^0$ decay and right, for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay.

As well as the overall hit rate, it is important to understand the rate in the

detectors produced by the daughter particles of the K_L decays. The daughter particles are selected as particles with a GEANT4 production process of "Decay", which should include any subsequent decays of the daughter particles, e.g. $\pi^0 \rightarrow \gamma\gamma$. This excludes all hits from particles produced by "gamma conversion", which produce electron-positron pairs and impact the rate, as described above. The rates for each detector are shown in Table 6. The overall rates are lower when compared to the total rates in Table 5, and the rates in the spectrometer for the $K_L \rightarrow \pi^0\pi^0\pi^0$ decay are much lower which is expected. Overall, the rates are acceptable for the detectors as it is assumed that most of the hits not produced from the decay of the K_L or other subsequent decays would be removed by the trigger system as they would be outside the time window of the K_L decay.

Detector	K_L Decay Particle Rates [MHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0\pi^0\pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+\pi^-\pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	14.7	11.2	1.3	5.6
STRAW-2	15.4	11.7	2.1	6.2
STRAW-3	16.1	12.1	2.5	6.6
STRAW-4	16.5	12.2	3.4	6.9
NCHOD	19.9	10.8	9.4	7.0
CHOD	16.1	11.3	8.1	7.8
IRC	1.9	0.8	1.8	1.0
LKr	15.4	11.6	19.9	10.5
MUV1	7.6	9.8	0.3	4.7
MUV2	2.7	7.2	0.1	1.5
MUV3	1.1	6.7	0.3	0.9
HAC	0.6	1.0	0.1	0.3

Table 6: Detector rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0\pi^0\pi^0$ and $K_L \rightarrow \pi^+\pi^-\pi^0$, produced directly from the daughter particles of the K_L decays.

It is important to understand the rates of the detectors in order to determine the overall detector design, especially for the electronics that would need to be designed

to meet the high rates. The reported rates would also drive the development of the overall experiment data acquisition and trigger systems. Along with the overall rate, it is vital to know the max channel rates in the detectors to design the electronics to handle the highest expected rate. The max channel rates for each detector are collected and are shown in Table 7.

Detector	K_L Decay Daughter Max Channel Rates [kHz]			
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 $\pm$ 0.11)%	$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 $\pm$ 0.07)%	$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 $\pm$ 0.12)%	$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 $\pm$ 0.05)%
STRAW-1	325	254	51	186
STRAW-2	377	254	27	154
STRAW-3	329	265	19	138
STRAW-4	365	258	39	138
NCHOD	484	329	186	206
CHOD	1290	635	786	559
IRC	1020	345	1063	516
LKr	575	321	385	293
MUV1	1830	1496	19	1032
MUV2	329	583	15	190
MUV3	79	234	39	43
HAC	206	210	27	138

Table 7: Max channel rates for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

As detailed previously in this section, the detectors which had finalised designs are implemented in the simulation, which allowed for correct rate calculation for all of these detectors. Other detectors that needed to be upgraded to increase their granularity, were not yet finalised and, therefore, not implemented in the simulation. This would result in an incorrect calculation as the rate is determined by the NA62 detector design and the layout of the channels in the detectors. However, it is still possible to calculate the rates at the planes of the upgraded detectors by collecting particles passing through the Z of the detector. This is possible using raw MC

data, where information for particles is stored at different parts of the HIKE setup (checkpoints). The rates for charged particles produced from the four K_L decays at each checkpoint are shown in Table 8.

Checkpoints (Z Position)	K_L Decay Particle Rates [kHz/cm ²]							
	$K_L \rightarrow \pi^\pm e^\mp \nu_e$ (40.55 ± 0.11)%		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ (27.04 ± 0.07)%		$K_L \rightarrow \pi^0 \pi^0 \pi^0$ (19.52 ± 0.12)%		$K_L \rightarrow \pi^+ \pi^- \pi^0$ (12.54 ± 0.05)%	
	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	19.28	0.34	16.64	0.25	0.05	10 ⁻³	10.55	0.13
STRAW-3 Entry (230.5 m)	6.93	0.29	6.23	0.23	0.14	0.02	4.04	0.12
NCHOD Entry (237.6 m)	4.93	0.16	4.43	0.13	0.17	0.01	2.66	0.06
CHOD Entry (238.9 m)	4.60	0.27	4.07	0.21	0.14	0.02	2.55	0.11
IRC Entry (239.5 m)	4.77	0.26	3.89	0.21	0.13	0.02	2.43	0.11
LKr Entry (240.4 m)	4.60	0.15	3.72	0.13	0.10	0.01	2.34	0.06
MUV1 Entry (243.2 m)	3.92	0.02	3.38	0.08	0.03	10 ⁻⁴	1.96	0.02
MUV2 Entry (244.4 m)	3.70	0.01	3.25	0.08	0.03	10 ⁻⁴	1.82	0.01
MUV3 Entry (245.3 m)	0.15	0.08	1.05	0.08	0.01	10 ⁻⁷	0.09	0.01
HAC Entry (253.3 m)	3.40	0.09	2.88	0.09	0.02	10 ⁻⁵	1.44	0.01

Table 8: Particle rates at specified checkpoints along the HIKE detector setup. Including the average rate across the cross-section and the maximum rate per cm^2 for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

However, as the rates are collected from the cross-section at the expected detector position, without accounting for the beam hole to accommodate beam particles passing through the detector the rates are higher than expected. An example cross-section at the entry of the first STRAW chamber is shown in Figure 29, where it is shown that the highest rate from the decay particles is found at the centre. As the detectors are not designed and the beam hole size is unknown, it is possible to estimate the size by calculating the beam divergence at the detector Z position. This is calculated using Equation 0.1.

$$\delta_{beam} = 0.4 \text{ mrad} \times Z \quad (0.1)$$

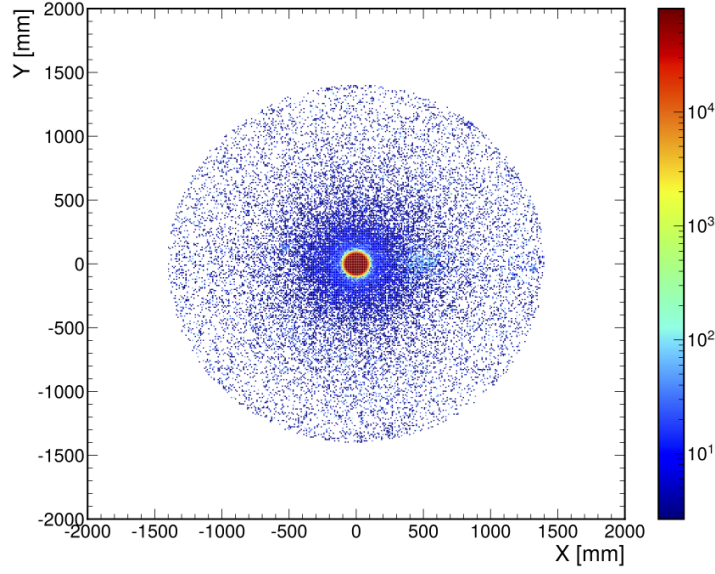


Figure 29: Rate of particles across the cross section at the first STRAW chamber entrance at $Z=213$. m.

Using Equation 0.1, a mask simulating the beam hole is applied to remove any particles that would pass through the detector. The resultant rates with the beam hole mask applied are shown in Table 9. As expected, this reduced the max rate for most detectors without impacting the average rate across the detector plane.

		K_L Decay Particle Rates [kHz/cm ²]							
		$K_L \rightarrow \pi^\pm e^\mp \nu_e$		$K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$		$K_L \rightarrow \pi^0 \pi^0 \pi^0$		$K_L \rightarrow \pi^+ \pi^- \pi^0$	
		(40.55 $\pm$ 0.11)%		(27.04 $\pm$ 0.07)%		(19.52 $\pm$ 0.12)%		(12.54 $\pm$ 0.05)%	
Checkpoints (Z Position)	Beam radius [mm]	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	85.4	11.59	0.305	9.504	0.226	0.047	0.001	5.863	0.109
STRAW-3 Entry (230.5 m)	92.2	5.515	0.280	4.900	0.219	0.139	0.017	3.087	0.113
NCHOD Entry (237.6 m)	95.0	3.959	0.152	3.143	0.124	0.166	0.012	2.032	0.063
CHOD Entry (238.9 m)	95.6	3.418	0.256	3.017	0.202	0.139	0.017	1.849	0.104
IRC Entry (239.5 m)	95.8	3.418	0.249	2.921	0.198	0.133	0.016	1.733	0.100
LKr Entry (240.4 m)	96.2	2.501	0.145	2.053	0.122	0.100	0.010	1.314	0.059
MUV1 Entry (243.2 m)	97.3	0.279	0.016	1.311	0.074	0.033	10 ⁻⁴	0.227	0.012
MUV2 Entry (244.4 m)	97.8	0.167	0.008	1.215	0.069	0.027	10 ⁻⁴	0.116	0.007
MUV3 Entry (245.3 m)	98.1	0.153	0.008	1.051	0.079	0.007	10 ⁻⁷	0.089	0.007
HAC Entry (253.3 m)	101.3	0.167	0.008	1.147	0.078	0.013	10 ⁻⁶	0.138	0.007

Table 9: Particle rates at specified checkpoints along the HIKE detector setup excluding particles passing within the expected beam radius for the four main K_L decays: $K_L \rightarrow \pi^\pm e^\mp \nu_e$ (K_{e3}^0), $K_L \rightarrow \pi^\pm \mu^\mp \nu_\mu$ ($K_{\mu 3}^0$), $K_L \rightarrow \pi^0 \pi^0 \pi^0$ and $K_L \rightarrow \pi^+ \pi^- \pi^0$, produced directly from the daughter particles of the K_L decays.

For the detector design, it is also important to calculate the total rates produced by different particles, which is shown in Table 10. The majority of the rates at the detector planes are from photons, especially in the spectrometer. The rate from charged particles suggests that the detectors could be designed and built with current technologies.

All rates information from the simulation shown in this section, would drive the design of the detectors that are not already upgraded. Table 10 is used in the HIKE proposal to demonstrate the requirement for upgrading the detectors to handle the increased rates. The particle rates are collected not from the geometrical acceptance of the corresponding sub-detectors but from the cross-section of the outer volume for each sub-detector found in the Monte Carlo simulation. For the Straw-1&3 the values of the cross-section used are, a circle of radius of 141.5 mm, NCHOD a square of length 330 mm, CHOD a square of length 244 mm, IRC a square of length 122 mm, MEC a circle of radius 182.5 mm, MUV-1&2 a square of length 320 mm, MUV-3 a square of length 281 mm and HAC a square of length 281 mm. A geometrical cut has been made to remove particles that pass through the checkpoints within the expected beam dispersion radius calculated to be $0.4 \times 10^{-3} \times Z$, where Z is the position of the checkpoint.

Checkpoints	Total Particle Rates kHz/cm ²									
	$e^{+/-}$		$\mu^{+/-}$		$\pi^{+/-}$		γ		Charged Particles	
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average
STRAW-1 Entry (213.5 m)	264	0.26	224	1.58	117	0.41	13426	13.87	360	2.56
STRAW-3 Entry (230.5 m)	246	0.59	179	1.34	120	0.40	11076	14.02	291	2.62
NCHOD Entry (237.6 m)	216	0.45	174	1.13	115	0.24	11770	8.50	238	1.98
CHOD Entry (238.9 m)	318	2.96	176	1.18	112	0.41	11687	15.57	376	5.79
IRC Entry (239.5 m)	251	3.85	176	1.17	85	0.40	11481	15.58	291	5.64
MEC Entry (240.4 m)	85	1.76	174	1.12	109	0.24	166	0.30	238	3.25
MUV1 Entry (243.2 m)	5	0.01	190	1.04	32	0.03	29	0.01	190	1.07
MUV2 Entry (244.4 m)	8	10^{-3}	162	1.00	5	10^{-3}	5	10^{-3}	163	0.98
MUV3 Entry (245.3 m)	5	10^{-4}	115	0.89	3	10^{-4}	3	10^{-3}	115	0.88
HAC Entry (253.3 m)	11	0.02	120	0.89	5	10^{-3}	3	10^{-3}	120	0.89

Table 10: Average and Max particle rates per cm^2 for specified checkpoints along the length of the detector.

0.6 Conclusions

Unfortunately, in 2024, the CERN management did not approve HIKE for operation in ECN3. However, the development of the detectors, software and the subsequent physics studies is currently being utilised for studies within the KOTO II proposal [20].

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